

Sentimental Analysis: A Comparative Study Over Different Social Media Platforms

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Abstract

An opinion mining which is also refereed as senti-Analysis is an essential techniques in Natural Language Processing that aims to identify emotions, opinions and attitudes within textual data. As social media continues to grow in popularity, platforms like Twitter, Facebook, Reddit and YouTube provides abundant resources for sentiment analysis featuring a variety of linguistic styles and user interactions. However, the structural and contextual distinctions among these platforms introduce the specific challenges such as difference in text length, sentiment conveying, slang usage, sarcasm and other certain domains. This study offers a comparative analysis of sentiment analysis across several social media platforms, focusing on the variations in data features, sentiment portrayal and the unique challenges faced by each platform. The main aim is to make the execution assessment measurements like accuracy, exactness, recall, F1- score up till tall as (92% – 98%). So here authors investigate the existing methods for sentiment analysis which includes lexicon-based, machine learning, deep learning strategies, assessing their effectiveness and constraints across diverse platforms.

Keywords: Twitter, Facebook, YouTube, Reddit, NLP, Sentiment, Deep Learning, Machine Learning, Analysis

1. Introduction

Social Media [1] has revolutionized communication for both individuals and organizations. Establishing itself as a vital source for public opinion, sentiment, and emerging trends. Platforms like twitter [14], Facebook [4], Reddit [8] and YouTube serve as digital forums where user express their views, participate in discussions and offer feedback on a wide array of subjects, including political issues and brand perceptions. The vast amount and variety of user-generated [9] content have led to the rise of sentiment analysis as a crucial method for deriving valuable insights from social media data. Companies utilize sentiment analysis to assess the customer satisfaction governments track public opinion regarding policies, and researchers who study online behaviors to identify new social trends.

However, the significance of sentiment analysis is accompanied by unique challenges across various social media platforms, stemming [6] from differences in content structure, linguistic intricacies, and user interaction styles. For instance, twitter's character limit results in brief, often unclear statements, while reddit features extensive discussions threads with sentiments that depend heavily on context. YouTube's comment sections are characterized by informal, conversational language, whereas Facebook [3] allows for longer and more structured posts that can include multimedia elements. These platform-specific characteristics influence the efficiency of sentiment analysis methods and necessitate customized strategies.

This survey intends to evaluate sentiment analysis[14] methodologies across various platforms by examine the lexicon-based, machine learning, and deep learning techniques [12] which focuses on their availability to access the different social media contexts. Addressing the significant challenges such as processing multilingual texts, dealing with slang, sarcasm, emoji's and various sentiment classifications [2, 3, 9]. Here authors investigate the availability of datasets, performance metrics and future research avenues to evaluate the current landscape of sentiment analysis [11, 13] and pinpoint areas for enhancement, including the utilization of large scale datasets and advance evaluation techniques.

This document examines sentiment analysis within social media platforms, focusing on the comparison of methodologies, challenges, and prospective developments. Section 2 evaluates the current body of research, whereas Section 3 outlines various sentiment analysis techniques. Section 4 presents a comparative analysis of sentiment analysis across different platforms, and Section 5 addresses the datasets that are accessible. Section 6 emphasizes significant challenges, Section 7 investigates avenues for future research, and Section 8 wraps up the study.

2. Literature Survey

Table 1. Findings of the literature survey

Author	Data Source	Pre-Processing Methods	Feature Extraction Methods	Classification Model	Performance metrics	Key Findings
[1] B.Vishnu Vardhan {1}-2024	Facebook (text, images)	Feature Extraction, Pre processing	Word2Vec, n-grams	BERT,CNN-LSTM, XINet, SVM	Accuracy, Precision	Implemented various extraction methods for a good sentiment analysis
[2] Umesh Chandra Jaiswal {13}-2023	Comments	Data extraction, Pre processing	Frequency analysis, external lexicon	Naïve Bayes, Logistic Regression, SVM	Accuracy, Precision, F1-score	Enhances sentiment classification through multiple ML algorithms
[3] Neha Shinde {12}-2024	Facebook, twitter, Blogs	Data Pre processing	Sentiment based, punctuation based features and	Naïve Bayes, Logistic Regression, Multilayer Perceptron	Accuracy, Precision, Recall, F1-score	Logistic Regression shows a high performance in practical applications
[4] Vishal.A Kharde {15}-2021	Twitter	URL, Hashtags, stop word removal	Affinity Analysis, pattern based	ML and Lexicon based	-	Address challenges like sarcasm, context sensitivity and domain adaption
[5] S. Veni {4}-2023	Facebook	Data Pre processing	Uni-gram, N-gram, TF-IDF	Naïve Bayes, SVM, Random Forest, KNN	Accuracy	SVM demonstrates the highest accuracy among all evaluated classifiers
[6]	E-	Informal	N-gram,	Naïve Bayes,	Accuracy,	SVM achieves

Palanichamy Naveen {2}-2024	Commerce Reviews	Language Handling	BOW, TF-IDF	Logistic Regression, SVM, LSTM	Precision, Recall, F1-score	the highest accuracy of 88%
[7] J.A. Serur {7}-2020	Reddit	Reddit comments gathered via PushFit API	TF-IDF, POS tagging	Naïve Bayes, Logistic Regression, CNN	Accuracy, Precision, Recall, F1-score	Correlates Reddit sentiment with stock prices
[8] Bhumika Gupta {14}-2022	Twitter	Removal of retweets, Stop word removal	Uni gram, Bi gram, TF-IDF	Naïve Bayes, Logistic Regression, SVM, ANN	Accuracy, Precision	Uses multiple feature extraction techniques and various classifiers
[9] Yudha Mahardika Putra {5}-2024	Instagram	Lower case punctuation, Lemmatization, tokenization	TF-IDF	SVM	Various execution assessment metrics	SVM achieves a correct rate of 89.41%
[10] Ang li Desouza {8}-2024	Reddit	Data Cleaning, tokenization, stop word removal	Spacy NLP, TF-IDF, custom classifier	Naïve Bayes, Logistic Regression, CNN	Accuracy, Precision, Recall	Correlates reddit sentiment with performance and uses advanced NLP

3. Overview of Sentiment Analysis

Analysis which plays a vital technique in language process [5] which focuses on evaluating and interpretation the emotional tone present in a text. This process entails determining whether the text expresses a positive, Negative or neutral sentiment. In more sophisticated application, it can achieve a detailed sentiment classification, distinguishing sentiments [7, 9] into various categories such as very negative. Given the rapid increase in content generated by any user across various platforms like forums and review sites and this has emerged [10] as an indispensable tool in numerous fields. So its primary application includes various phases as mentioned below.

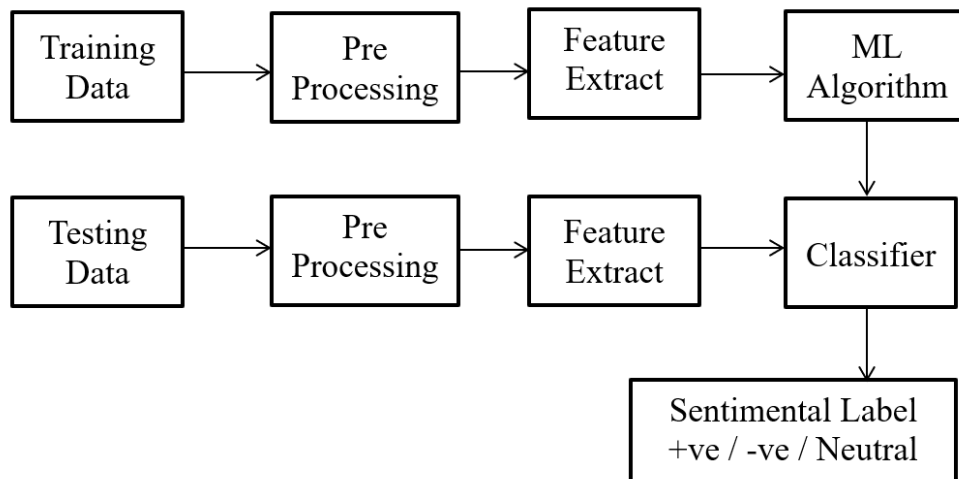


Fig. 1. Block Diagram

Product Review and Brand Monitoring:

Business Analyze customer feedback from online reviews to evaluate product performance, discern customer preferences and enhance their offerings. Additionally sentiment analysis [5] plays a crucial role in monitoring brand reputation, enabling companies to address consumer concerns effectively.

Social Media Monitoring and Crisis Management:

Organizations monitor brand mentions, Hashtags and conversations on social media to identify the emerging trends, evaluate customer engagement [1, 4] and manage crises. Sentiment analysis aids in recognizing negative shifts in public opinion allowing businesses or governments to take timely and proactive measures.

Customer Feedback and Service Improvement:

Companies employ sentiment analysis on customer service interactions, surveys and chatbot dialogues to pinpoint areas of concerns and enhance user experiences. By assessing customer sentiment in real time, organizations can refine product recommendations [9], customize advertisements and optimize services.

3.1 Approaches to Sentiment Analysis

Analysis strategies can be categorized into three essential sorts: lexicon based, ML strategies and profound based strategies. Each of these approaches presents special qualities and shortcoming, which are formed by different components counting the complexity of the content, so that the accessibility of datasets, computational capabilities and the need for relevant comprehension.

The more approaches have seen in the some sections and its working algorithm for implementation. So now the various approaches based on Deep Learning, Machine Learning, various Transformers based and hybrid approaches achieves a long consistency over different percentage rate on the dataset.

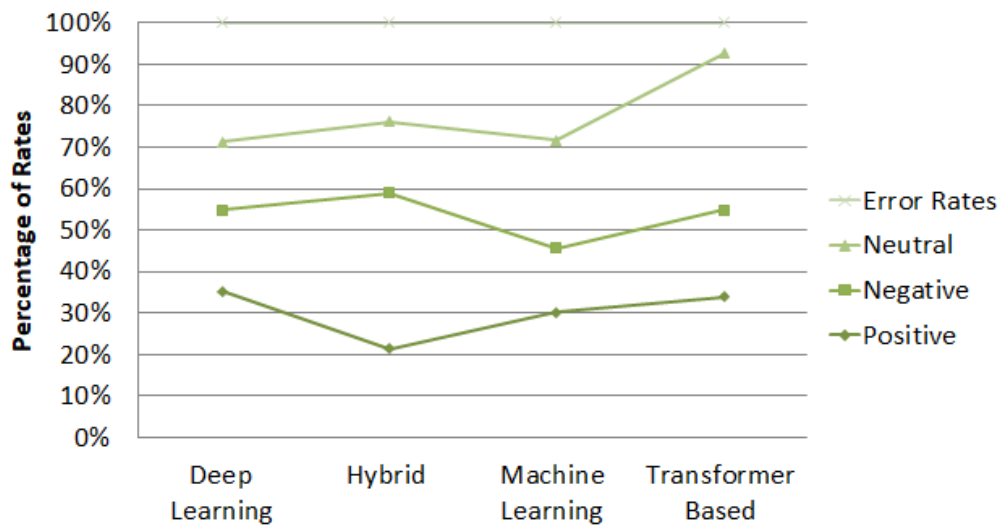


Fig. 2. Sentiment Analysis Approaches

3.1.1 Methods based on Lexicon type

Lexicon based methods utilize established sentiment lexicons, which are collections of words paired with sentiment scores. These scores reflect whether a word express a positive, negative or neutral Notable sentiment lexicon include:

SentiWordNet: A lexical database that assigns sentiment scores to words based on their definitions. VADER tailored for analyzing sentiment in social media[7, 11] content, adept at interpreting slang, emoji's and punctuation whereas a compilation of English words linked with sentiment scores, beneficial for straightforward sentiment classification tasks.

3.1.2 Methods based on ML

ML approaches entail training modeling on labeled datasets to identify patterns within the text and automatically classify sentiment. These models derive textual features such as word frequency, grams, and various frequency inverse document and various word embedding's to generate the various predictions. So some of the machine learning algorithms includes:

Naive Bayes: A probabilistic classifier grounded in Bayes theory, [14] commonly applied in text classification.

Support Vector Machines: A supervised learning technique that determines the optimal hyperplane for sentiment classification.

Random Forest: An gather learning procedure that improves classification exactness by combining different D-Trees.

Logistic Regression: The statistical model utilized for binary or multi-class sentiment classification tasks. Machine learning approaches typically demonstrated the superior performance compared to lexicon- based methods as they derive insights from patterns within labeled datasets instead of depending on established rules. Nonetheless, these methods necessitate high quality labeled data and may encounter the various challenges which is particularly has a across various social media platforms.

3.1.3 Deep learning techniques:

Deep learning methodologies have markedly enhanced sentiment classification [5] by using N- networks, identifies the intricate linguist formations and contextual meanings. These models take the advantage of extensive datasets and sophisticated architectures [3] to achieve greater accuracy. The most used models in field of AI are:

Recurrent Networks: Tailored for continuous data, making them adept at processing sentences and recognizing word dependencies.

Deep Memory Networks: A modification of RNN's designed to manage long-range dependencies in text, thereby enhancing sentiment classification accuracy.

Transformer Models: The two directional encoder which represents from (BERT) and similar frameworks utilize already trained contextual embedding's to interpret the sentiment in a more nuanced and précised manner. These models surpass traditional machine learning methods but demand substantial computational resources and extensive datasets for effective fine-tuning.

Hybrid Approaches: To address the shortcomings of singular methodologies, hybrid models integrate lexicon-based, ML and DL techniques to enhance the correctness of sentiment classification.

For instance, a lexicon-based method may serve for initial preprocessing and classification which is subsequently refined by a machine learning or deep learning model. These hybrid models significantly robustness, especially in intricate sentiment analysis scenarios.

3.2 Selecting the Appropriate Method

The selection of sentiment analysis techniques is contingent upon the specific application, the availability of datasets, computational limitations and performance expectations. Lexicon-based methods are efficient and easily interpretable [2, 6] yet they often lack the ability to grasp context.

Machine learning models provide superior accuracy at the cost of requiring substantial computational power. Hybrid approaches strike a balance between efficiency and accuracy, rendering them ideal for practical applications where varied text formats are prevalent across social media platforms. [12]

Lexicon-Based Method Steps:

Input the data

Apply the preprocessing

- Convert character to down case
- Delete non-alphanumeric symbols, URL's, stop words etc.

- Perform word normalization

Tokenization: Split the cleaned text into individual words

Load Sentiment Lexicon

Sentiment Scoring: Initialize the counters for +ve, -ve and neutral words

Aggregate score metrics: Calculate the sentiment proportions

Machine learning Method Steps:

Input the data

Apply the preprocessing

- Make characters to lower case
- Exclude special letters like URL's and stop words
- Process word roots

Extraction of vital features: Transform text into numbers like TF-IDF or BoW

Dataset Splitting & Model training: Split the data and choose models (SVM, Regression)

Model Evaluation & Prediction: Train the model and calculate metrics (accuracy, precision, F1-score, recall)

4. Comparison of Platforms

Social media platforms possess distinct data architectures, user engagement patterns and communication styles all of which profoundly influence sentiment analysis [5] techniques. The success of sentiment analysis is contingent upon the algorithms ability to accommodate these platforms specific differences such as variations of text length, the use of colloquial language, [10] sarcasm and methods of interaction. Presented below is a comparative examination of sentiment analysis methodologies utilized across prominent platforms.

Each platform necessitates tailored sentiment analysis methodologies that take into account its distinct features. The following is a comprehensive [8] examination of the challenges and strategies associated with sentiment analysis for each platform.

Table 2. Comparison over Social Platform

Platform	Data Characteristics	Challenges
Twitter	Short text (280 characters), hashtags, emoji's, trending Topics	Sarcasm, abbreviations, slang, character limit
Facebook	Longer posts, reactions, comments, user interactions	Spam, misinformation, fake news
Reddit	Threaded discussions, community voting	Context dependent sentiment, nested replies, diverse topics
YouTube	Video comments, informal language, highly engaged Discussions	Noisy data, bot generated comments, sarcasm, lack of context

4.1 Twitter

Data characteristics:

Twitter restricts posts to 280 characters, promoting brevity and a causal tone. The relevant use of hashtags, emoji's, slang and various abbreviations and the interpretation of sentiment. [14, 15] The platform's real time dynamics contribute to a constantly changing landscape of language and sentiment trends.

Challenges:

- **Sarcasm Detection:** The brevity of tweets often includes sarcasm which poses challenges for traditional

lexicon based or machine learning models to accurately interpret.

- **Evolving trends:** The meanings of hashtags frequently shift, necessitating that sentiment models adapt in real time.

Common Techniques:

- Methods based on lexicon types
- Methods based on ML
- Methods based on DL

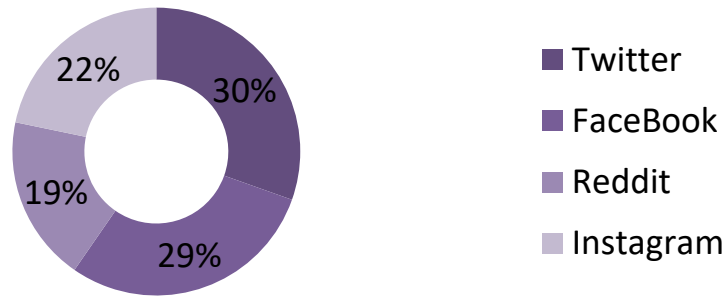


Fig. 3. Percentage of Social Media

4.2 Facebook

Data characteristics:

In contrast to Twitter, Facebook accommodates long form content, including status updates, articles and discussions. The platform's rich engagement features [3] such as likes, reactions, shares and comments which provides the additional cues for sentiment analysis.

It also consists of various phases like preprocessing of various tasks in the context of natural language processing and other analysis. In the context of sentiment [4] of various happy, sad, angry the challenges are listed below to get better understanding.

Challenges:

- **Misinformation & fake News:** Sentiment analysis model faces the challenge of distinguishing between the factual information and emotional responses.
- **Inconsistent formatting:** The presence of images, videos and links in Facebook posts complicates the text extraction and preprocessing.

Common Techniques:

- Text & Metadata based models
- Deep learning for Sentiment prediction

4.3 Reddit

Data characteristics:

Reddit features threaded discussions where sentiments [8] develop as users interact with one another. The community driven ranking of comments is facilitated by up-votes and down-votes. The discussion encompasses a wide range of topics from political issues to specialized [7] interests which is necessary for domain adaptation for effective sentiment modeling.

Challenges:

- **Contextual Sentiment:** A single discussion thread may present conflicting opinions complicating the assignment of a singular sentiment label.
- **Mixed Sentiment within Threads:** Users may express a positive sentiment initially only to alter their views based on subsequent replies.

Common Techniques:

- Topic modeling
- Hierarchical Sentiment Analysis

4.4 YouTube

Data characteristics:

Sentiment analysis on YouTube primarily targets video comments, which tend to be informal, highly interactive and relined on context. Comments may vary in length [4] from brief remarks to extensive discussions for effective analysis.

Challenges:

- **Noisy Data:** Numerous comments feature misspellings, emoji's and internet slang, making thorough text preprocessing crucial.
- **Lack of Context:** the sentiments expressed in comments frequently relate to the content of the video which requires a multimodal approach to sentiment analysis.

Common Techniques:

- ML and NLP models
- Multimodal Sentiment analysis

5. Dataset Description for YouTube Comments

A YouTube Comments Dataset typically includes text comments posted on YouTube videos, along with metadata related to user engagement and video context. This dataset can be used for sentiment analysis, toxicity detection, engagement analysis, and other NLP-based research. The 1st Table shows about the Dataset which was used for the coding the implementation of the model and the 2nd Table shows actual; keywords extracted. So these examples are given below in terms of implementation and its results:

(The link for the implemented code: <https://github.com/SahilWarade/YouTube-Comments-Sentiment-Analysis>)

Table 3. Dataset example

Sr. No.	YouTube Comments Dataset Examples
1	Please make a video on which exam to be given to get into foreign universities and colleges and detailed criteria. As I have recently given 10 boards I want to know more about what to do from here and how to start coding ♥
2	You all need to crack IELTS exam to get admission into any college or university where english is a first language and I've been teaching it from past 1 year please contact for details and enroll in my online & offline classes to crack ielts in your first attempt.
3	Hi, my son has got admission for undergraduate in engineering mechanics in UIUC and Data science in UCSD. Which factors should he look at while finalizing college? Please can you comment on this?
4	Anyone for btech abroad? Pls reply
5	It's so confusing to find an reputed University across the country!!
6	Chalo Jaldi jaldi Germany ja ke Settle ho jata hu....
7	Kindly correct your info on US - you get only 90 days to find a job after masters.

Extracted Output keywords for evaluating the models were:

Table 4. Extracted Keywords

Sr. No.	Extracted and Matched Keywords for Model Prediction
1	'accused': 'cheated'

2	'acquitted': 'singled out'
3	'adorable': 'loved'
4	'adored': 'loved'
5	'affected': 'attracted'
6	'afflicted': 'sad'
7	'aghost': 'fearful'
8	'agog': 'attracted'
9	'agonized': 'sad'

Predicted Outcome Graph is shown in Fig. 4.

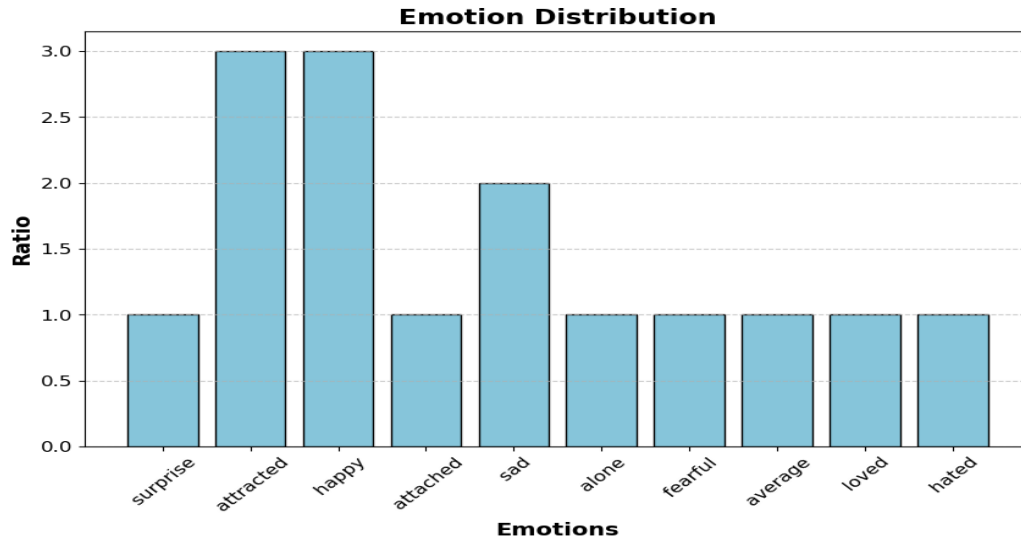


Fig. 4. Model Outcome

Table 5. Key Metrics

Accuracy	Precision	Recall	F1-Score
90%	80%	82%	75%

5.1 Key Takeaways

- The show is making redress forecasts most of the time, showing solid overall performance.
- When system predicts a certain emotion, it is **mostly correct**. This suggests **low false positives**, meaning it is cautious about making incorrect predictions.
- The model is **correctly identifying most of the actual emotion instances**. This means **low false negatives**, so it captures emotions well across different categories.
- This could mean **an imbalance in predictions across classes**, where some emotions are classified well but others poorly.

5.2 Publicly widely used datasets

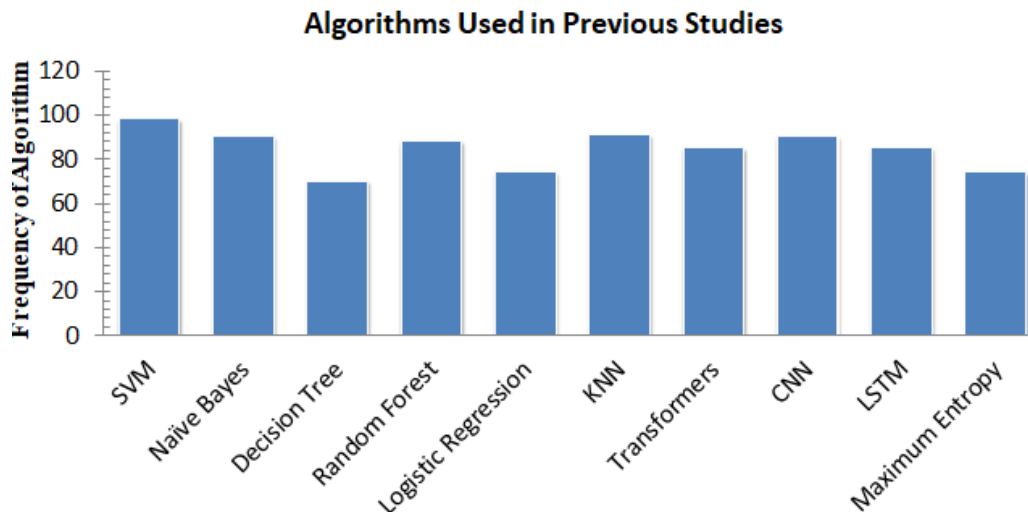
There are various datasets exists for training and evaluating sentiment analysis models which covers various types of platforms which are mention below the famous datasets:

Table 6. List of Datasets

Dataset	Platform	Description
Sentiment140	Twitter	1.6M tweets labeled as +ve / -ve
TweetEval	Twitter	Benchmark for sentiment, emotion and hate speech detection
Reddit Sentiment Dataset	Reddit	Subreddit specific sentiment labeled comments
YouTube Comments Dataset	YouTube	Video comments labeled with sentiment
IMDB Movie Reviews Dataset	General	50K movie reviews labeled as +ve/-ve
Amazon Reviews Dataset	General	Millions of reviews with star ratings (1 – 5)

6. Challenges in Sentiment Analysis

- **Text complexity:** Social media communications [6] frequently utilize the casual language, abbreviations, emoji's and colloquialisms which complicates the accurate interpretation of sentiment by NLP models.
- **Sarcasm and Irony:** The sentiment conveyed through sarcasm, irony and humor poses significant challenges for detection by rule based and standard machine learning methods [9] as the literal interpretation of words.
- **Multilingual Text & Code Mixing:** A considerable number of users engage in code mixing so thereby complicating sentiment classification for models [11] that are trained on models.
- **Noisy Data:** The presence of spam comments, fraudulent accounts, advertisements and content generated by the bots introduces extraneous noise into datasets which diminishes the precision of models.
- **Evolving language trends:** The language used on social media is emerging regularly, rendering traditional sentiment models which increases the pace [13] with these developments.

**Fig. 5.** Comparison of Algorithms

7. Future Directions

Significant advancements have been achieved in sentiment analysis on important media platforms such as twitter, Facebook & reddit. However several emerging offers distinct challenges and opportunities for research in sentiment analysis:

- **TikTok:** As a platform primarily centered on video content, TikTok presents difficulties for conventional text-based sentiment analysis. Research efforts must focus on multimodal analysis which integrates the visual and auditory elements to effectively access the user sentiment.
- **ClubHouse:** This audio platform facilitates live discussions complicating the capture and analysis of sentiment due to the ephemeral nature of conversations. Developing techniques for real-time audio data processing and analysis could yield valuable insights into user sentiment on such platforms.
- **Discord:** Mainly utilized by gaming communities, Discord features both real time text and voice chats. The informal language use of slang and prevalence of voice communications introduce unique challenges for sentiment analysis.
- **Telegram:** With its emphasis on privacy and encrypted messaging, accessing and analyzing data from telegram poses difficulties. Nevertheless, public channels and groups present opportunities for sentiment analysis particularly in regions where the telegram enjoys the widespread use.

8. Conclusion

This study provides a comparative analysis of sentiment analysis methodologies utilized across various social media platforms, emphasizing the distinctions in data structures, modes of sentiment expression and challenges unique to each platform. It is been examined prevalent lexicon based, machine learning and deep learning techniques which evaluates their accuracy on sentiment classification on twitter, Facebook and YouTube as well as reddit.

Although the machine learning and deep learning frameworks have markedly improved the accuracy of sentiment analysis, enduring the issues such as the detection of sarcasm, processing of multilingual texts, handling of noisy data etc. The accuracy of 90%, precision 80%, F1-Score 75% and recall 82% is achieved for the implemented code. So, the Future investigations should prioritize cross platform sentiment analysis, allowing models to adapt and generalize across diverse social media environments. Additionally the development of multimodal sentiment analysis, which combines text, audio, images and video is essential for platforms like TikTok, ClubHouse and YouTube where the sentiment is often expressed beyond more text.

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